

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	N. PARNITA	Reg. No.:	192525322
Section / Batch:	2025 - 2029	Date:	8.8.26

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1

SECTION A – Technical Presentation

Topic: Selection, Bubble & Insertion Sort :
 A comparative Performance Study

Code of Conduct

I N. Parnita (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

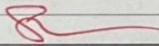
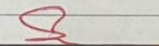
Signature of the Student: N. Parnita

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30%	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25%	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20%	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15%	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10%	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30%				
Problem Identification	25%				
Use of Tools	20%				
Communication	15%				
Professionalism	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: 	Signature: _____	Signature: 
Date: <u>3/8/20</u>	Date: _____	Date: <u>3/8/20</u>

CSA0617 – DESIGN AND ANALYSIS OF ALGORITHMS

Co1- Assessment –3

Reg No:192525322

Technical Presentation

Name: N. PARNITA

Topic: Selection, Bubble & Insertion Sort: A Comparative Performance Study



Selection, Bubble, and Insertion Sort are simple comparison-based sorting algorithms used to arrange data efficiently. Although all three have a worst-case time complexity of $O(N^2)$, their efficiency differs depending on the input data. **Insertion Sort** performs best for nearly sorted data; **Selection Sort** uses fewer swaps, and **Bubble Sort** is simple but generally less efficient for large datasets.

Key Efficiency Metrics

- **Time Complexity:** $O(N^2)$ across all three algorithms in the Average and Worst cases, while Insertion Sort achieves $O(N)$ in the best case for nearly sorted data.
- **Space Complexity:** $O(1)$ auxiliary space for all three algorithms, as they sort the array in place without requiring additional memory.
- **No. of Swaps:** Selection Sort performs fewest swaps, Bubble Sort performs the highest no. of swaps, & Insertion Sort shifts elements instead of performing frequent swaps.
- **Stability:** Bubble and Insertion Sort are **stable**, preserving the relative order of equal elements, whereas Selection Sort is **unstable** & may change their order during swapping.

N. Parnita

Signature